

Unsupervised Learning - K Means Clustering and PCA

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Set working Directory

```
# setwd("C:/Users/user/Desktop/Professional DS Mule/R")
```

Task 1 : Data Import and Exploration

Load required libraries

```
library(tidyverse)      # Data Manipulation
```

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.2.0      ✓ readr      2.2.0
## ✓ forcats    1.0.1      ✓ stringr    1.6.0
## ✓ ggplot2    4.0.3      ✓ tibble     3.3.1
## ✓ lubridate  1.9.5      ✓ tidyr      1.3.2
## ✓ purrr      1.2.1
## — Conflicts ————— tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(cluster)        # Clustering Algorithms
library(factoextra)     # Clustering Visualization
```

```
## Welcome to factoextra!
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practical-guide-to-principal-component-methods-in-r/
```

```
library(palmerpenguins) # Sample Dataset
```

```
##
## Attaching package: 'palmerpenguins'
##
## The following objects are masked from 'package:datasets':
##
##     penguins, penguins_raw
```

```
library(dplyr)          # Data Manipulation
library(ggplot2)        # Data Visualization
```

Load Penguin Dataset

```
data("penguins", package = "palmerpenguins")
head(penguins)
```

```
## # A tibble: 6 × 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm
##   <fct>   <fct>          <dbl>         <dbl>         <int>
## 1 Adelie  Torgersen         39.1          18.7           181
## 2 Adelie  Torgersen         39.5          17.4           186
## 3 Adelie  Torgersen         40.3           18           195
## 4 Adelie  Torgersen          NA           NA             NA
## 5 Adelie  Torgersen         36.7          19.3           193
## 6 Adelie  Torgersen         39.3          20.6           190
## # i 2 more variables: sex <fct>, year <int>
```

Explore the dataset

```
library(psych)
```

```
##
## Attaching package: 'psych'
```

```
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
str(penguins)
```

```
## tibble [344 × 8] (S3: tbl_df/tbl/data.frame)
## $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1
1 1 1 1 1 1 1 1 1 1 ...
## $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3
3 3 3 3 3 3 3 ...
## $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9
39.2 34.1 42 ...
## $ bill_depth_mm : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 1
9.6 18.1 20.2 ...
## $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 19
3 190 ...
## $ body_mass_g   : int [1:344] 3750 3800 3250 NA 3450 3650 3625
4675 3475 4250 ...
## $ sex          : Factor w/ 2 levels "female","male": 2 1 1 NA
1 2 1 2 NA NA ...
## $ year         : int [1:344] 2007 2007 2007 2007 2007 2007 200
7 2007 2007 2007 ...
```

```
describe(penguins)
```

```
##               vars   n   mean    sd median trimmed   mad
min      max
## species*         1 344    1.92  0.89    2.00    1.90    1.48
1.0     3.0
## island*          2 344    1.66  0.73    2.00    1.58    1.48
1.0     3.0
## bill_length_mm   3 342   43.92  5.46   44.45   43.91    7.04
32.1    59.6
## bill_depth_mm    4 342   17.15  1.97   17.30   17.17    2.22
13.1    21.5
## flipper_length_mm 5 342  200.92 14.06  197.00  200.34   16.31   1
72.0   231.0
## body_mass_g      6 342 4201.75 801.95 4050.00 4154.01  889.56  27
00.0 6300.0
## sex*             7 333    1.50  0.50    2.00    1.51    0.00
1.0     2.0
## year             8 344 2008.03  0.82 2008.00 2008.04    1.48  20
07.0 2009.0
##               range skew kurtosis    se
## species*         2.0  0.16   -1.73  0.05
## island*          2.0  0.61   -0.91  0.04
## bill_length_mm   27.5  0.05   -0.89  0.30
## bill_depth_mm    8.4 -0.14   -0.92  0.11
## flipper_length_mm 59.0  0.34   -1.00  0.76
## body_mass_g     3600.0  0.47   -0.74 43.36
## sex*             1.0 -0.02   -2.01  0.03
## year             2.0 -0.05   -1.51  0.04
```

Check for missing values

```
# Count missing values per column
missing_count <- colSums(is.na(penguins))
missing_count
```

```
##           species            island  bill_length_mm  bill_dept
h_mm
##           0              0              2
2
## flipper_length_mm  body_mass_g          sex
year
##           2              2              11
0
```

```
# percentage of missing values
sum(is.na(penguins)) / (nrow(penguins) * ncol(penguins)) * 100
```

```
## [1] 0.690407
```

remove rows with missing values

```
penguins_clean <- na.omit(penguins)
dim(penguins_clean)
```

```
## [1] 333    8
```

Task 2 :Data Preprocessing

```
# Select only numeric columns for clustering
numeric_vars <- c("bill_length_mm", "bill_depth_mm", "flipper_length_m
m", "body_mass_g")
penguins_numeric <- dplyr::select(penguins_clean, all_of(numeric_vars))
```

```
# Scale variables for clustering
penguins_scaled <- as.data.frame(scale(penguins_numeric))
```

```
# Verify Scaling
colMeans(penguins_scaled) # Should be close to 0
```

```
##      bill_length_mm      bill_depth_mm flipper_length_mm      body_ma
ss_g
##      3.552797e-16      5.572578e-16      1.834379e-16      -9.274780
e-17
```

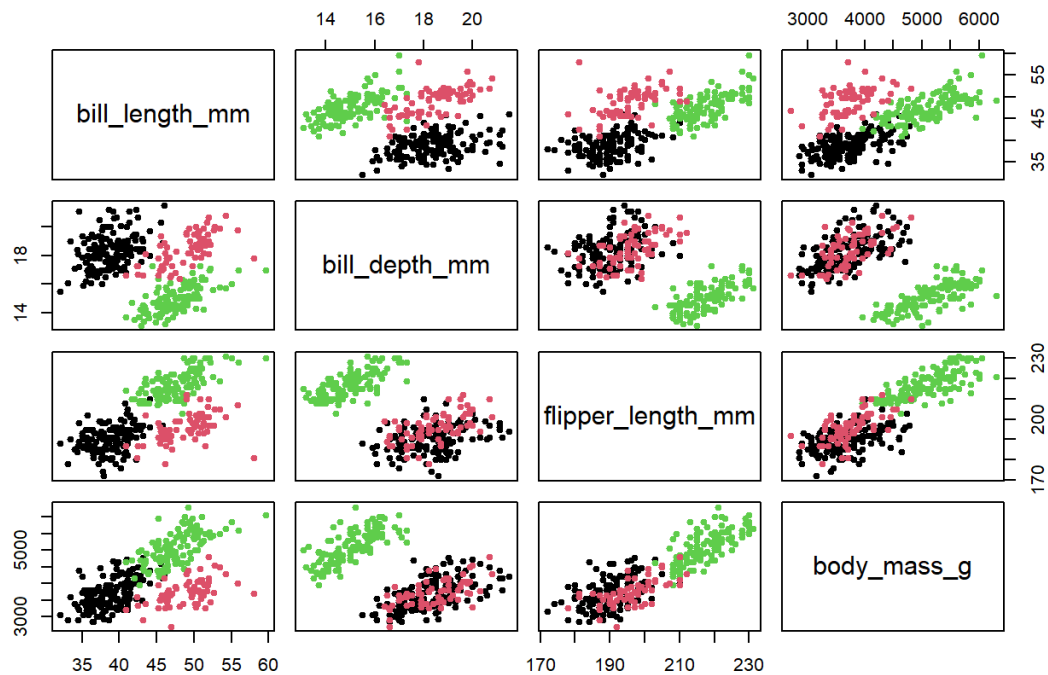
```
apply(penguins_scaled, 2, sd) # Should be close to 1
```

```
##      bill_length_mm      bill_depth_mm flipper_length_mm      body_ma
ss_g
##              1              1              1
1
```

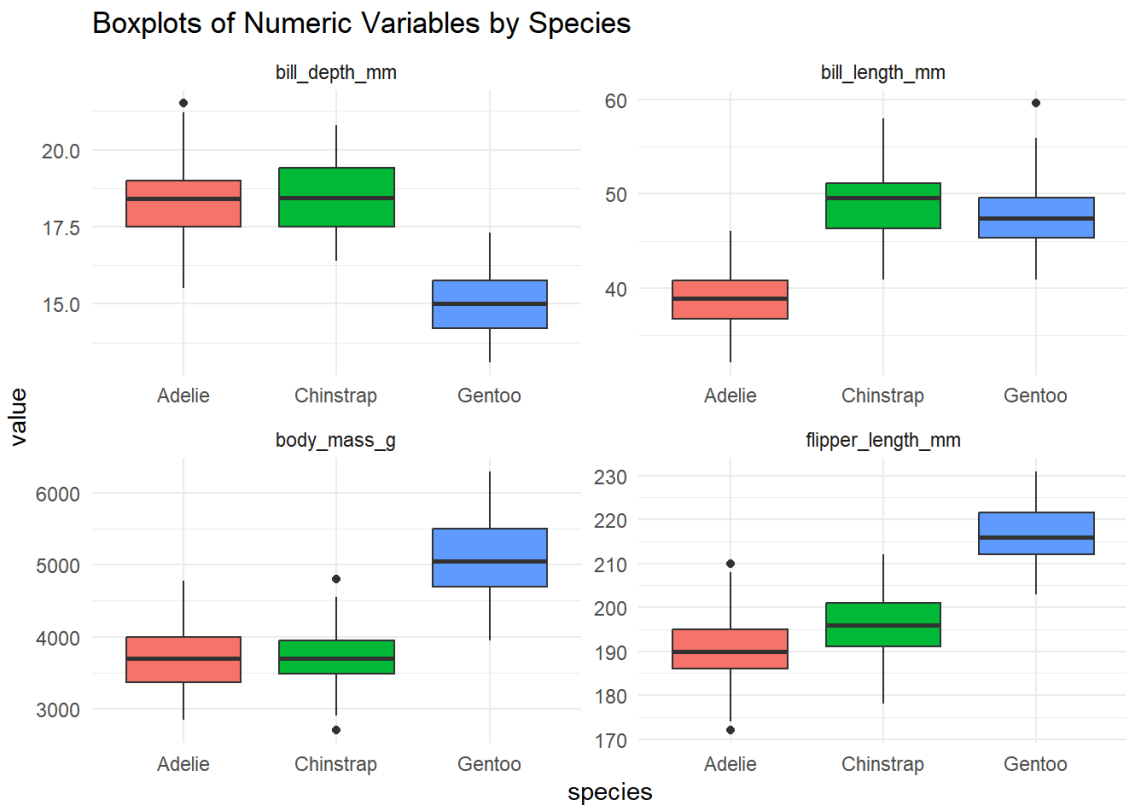
Task 3 : Exploratory Data Analysis

```
# Pairs plot colored by species
pairs(penguins_numeric, col = penguins_clean$species, pch = 19, cex =
0.7, main = "Penguin Morphology : Pairs Plot")
```

Penguin Morphology : Pairs Plot



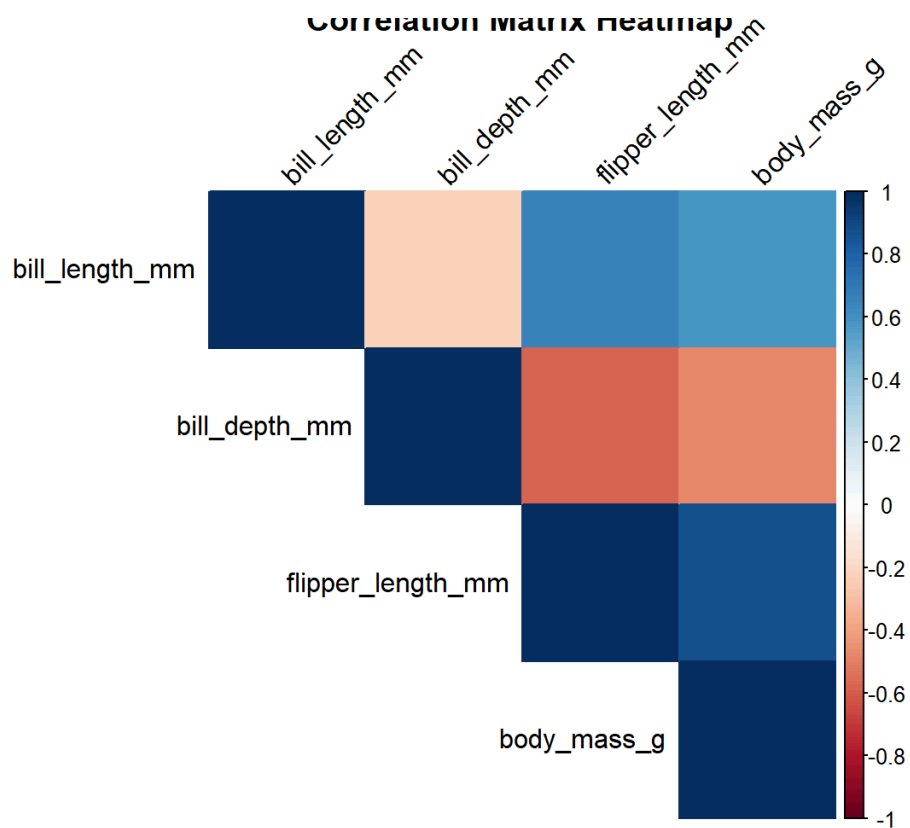
```
# Boxplots for each numeric variable by species
penguins_clean %>%
  pivot_longer(cols = all_of(numeric_vars), names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = species, y = value, fill = species)) +
  geom_boxplot() +
  facet_wrap(~ variable, scales = "free") +
  labs(title = "Boxplots of Numeric Variables by Species") +
  theme_minimal() +
  theme(legend.position = "none")
```



```
# Correlation matrix and heatmap
cor_matrix <- cor(penguins_numeric)
cor_matrix
```

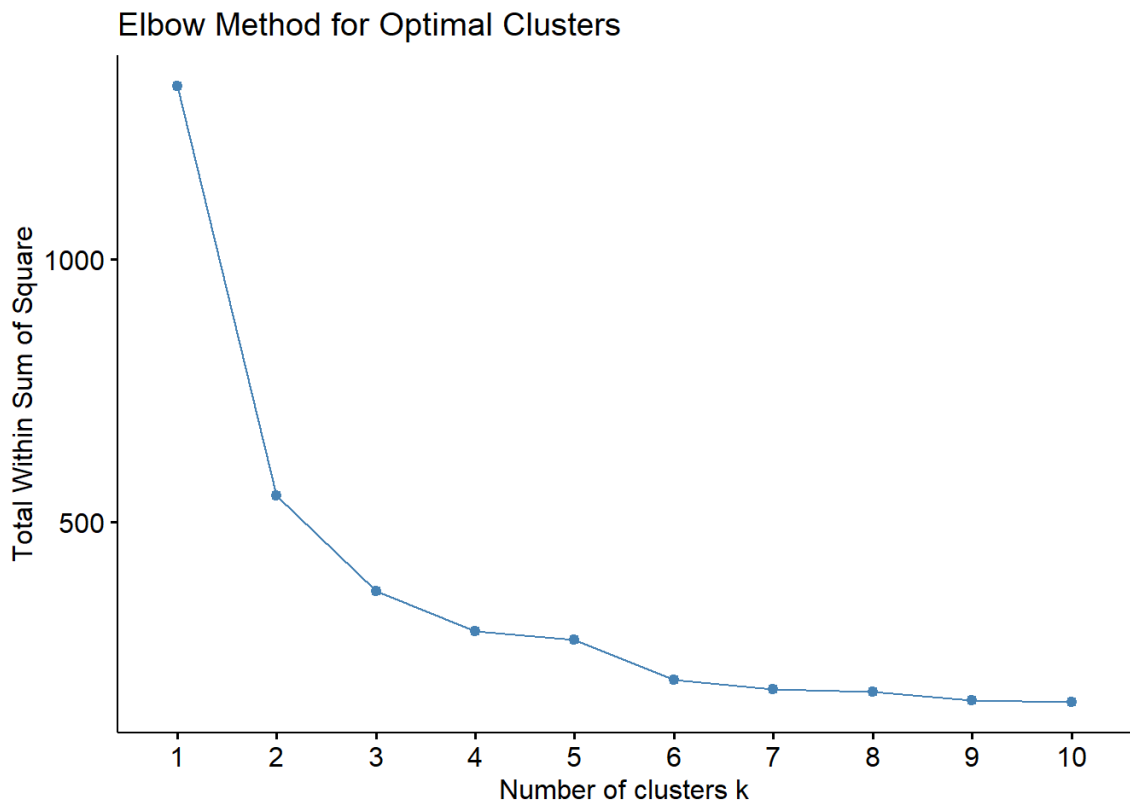
```
##               bill_length_mm bill_depth_mm flipper_length_mm bo
dy_mass_g
## bill_length_mm      1.0000000    -0.2286256      0.6530956
0.5894511
## bill_depth_mm      -0.2286256      1.0000000     -0.5777917  -
0.4720157
## flipper_length_mm   0.6530956     -0.5777917      1.0000000
0.8729789
## body_mass_g         0.5894511     -0.4720157      0.8729789
1.0000000
```

```
corrplot::corrplot(cor_matrix, method = "color", type = "upper", tl.co
l = "black", tl.srt = 45, title = "Correlation Matrix Heatmap")
```

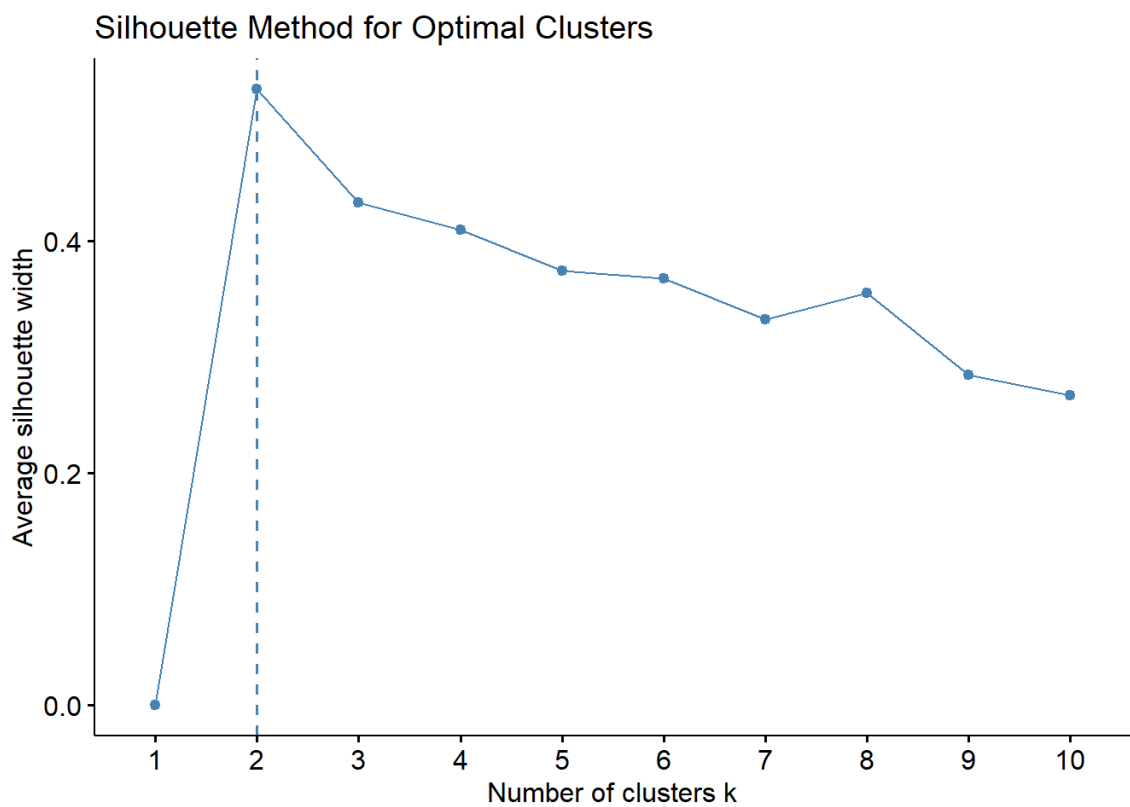


Task 4 : K-Means Clustering

```
set.seed(2026) # For reproducibility
# Determine optimal number of clusters using Elbow Method
fviz_nbclust(penguins_scaled, kmeans, method = "wss") +
  labs(title = "Elbow Method for Optimal Clusters")
```

```
# Determine optimal number of clusters using Silhouette Method  
fviz_nbclust(penguins_scaled, kmeans, method = "silhouette") +  
  labs(title = "Silhouette Method for Optimal Clusters")
```



From the Elbow and Silhouette methods, we can see that the optimal number of clusters is 2.

Task 5 : K-Means : Model Fitting and Cluster Profiling

```
set.seed(2026) # For reproducibility
k_opt <- 2
# Fit K-Means model with 3 clusters
kmeans_model <- kmeans(penguins_scaled, centers = k_opt, nstart = 25)
kmeans_model
```



```
# Cluster profiling
cluster_profiles <- penguins_clean %>%
  group_by(Cluster) %>%
  summarise(across(where(is.numeric), mean, na.rm = TRUE), n = n())
```

```
## Warning: There was 1 warning in `summarise()`.
## i In argument: `across(where(is.numeric), mean, na.rm = TRUE)`.
## i In group 1: `Cluster = 1`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
## # Previously
##   across(a:b, mean, na.rm = TRUE)
##
## # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

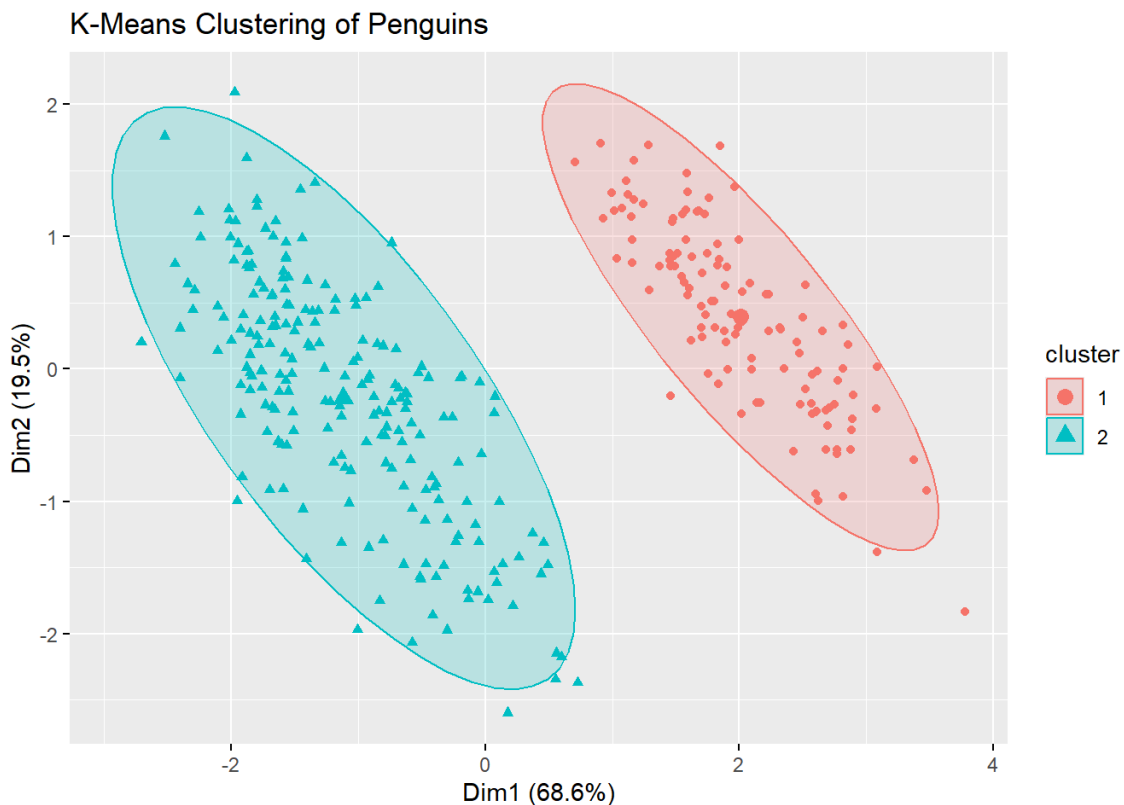
```
cluster_profiles
```

```
## # A tibble: 2 × 7
##   Cluster bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <dbl>      <dbl>         <dbl>         <dbl>         <dbl>
## 1 1          47.6           15.0           217.          509
## 2 2          42.0           18.4           192.          371
```

```
# Table summarizing cluster characteristics(size,centroid values,dominant species)
cluster_summary <- penguins_clean %>%
  group_by(Cluster) %>%
  summarise(Size = n(),
            Centroid_Bill_Length = mean(bill_length_mm),
            Centroid_Bill_Depth = mean(bill_depth_mm),
            Centroid_Flipper_Length = mean(flipper_length_mm),
            Centroid_Body_Mass = mean(body_mass_g),
            Dominant_Species = names(sort(table(species), decreasing = TRUE)[1]))
cluster_summary
```

```
## # A tibble: 2 × 7
##   Cluster Size Centroid_Bill_Length Centroid_Bill_Depth Centroid_F
lipper_Length
##   <fct>    <int>                <dbl>                <dbl>
<dbl>
## 1 1          119                47.6                15.0
217.
## 2 2          214                42.0                18.4
192.
## # i 2 more variables: Centroid_Body_Mass <dbl>, Dominant_Species <chr>
```

```
# Visualization
fviz_cluster(kmeans_model, data = penguins_scaled, geom = "point", ellipse.type = "norm", main = "K-Means Clustering of Penguins")
```

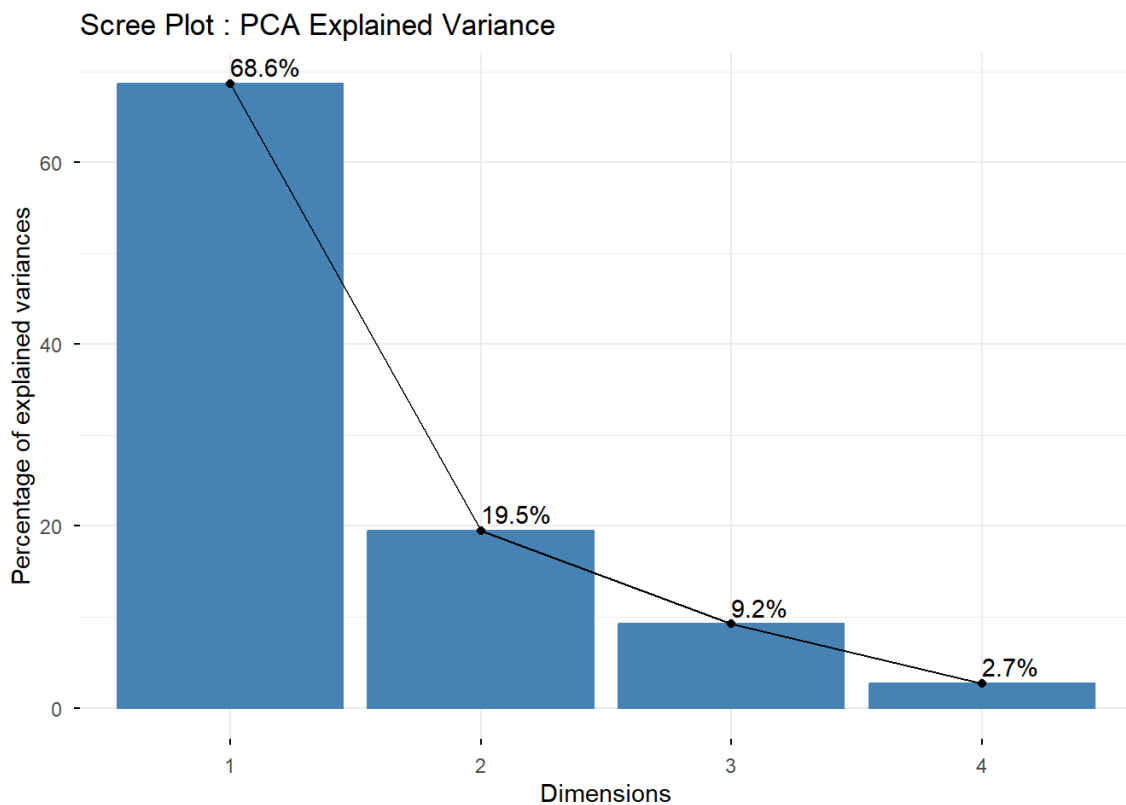


Task 6 : Principal Component Analysis (PCA)

```
# Perform PCA
pca_result <- prcomp(penguins_scaled, center = TRUE, scale. = TRUE)
# Summary of PCA
summary(pca_result)
```

```
## Importance of components:
##               PC1      PC2      PC3      PC4
## Standard deviation  1.6569 0.8821 0.60716 0.32846
## Proportion of Variance 0.6863 0.1945 0.09216 0.02697
## Cumulative Proportion 0.6863 0.8809 0.97303 1.00000
```

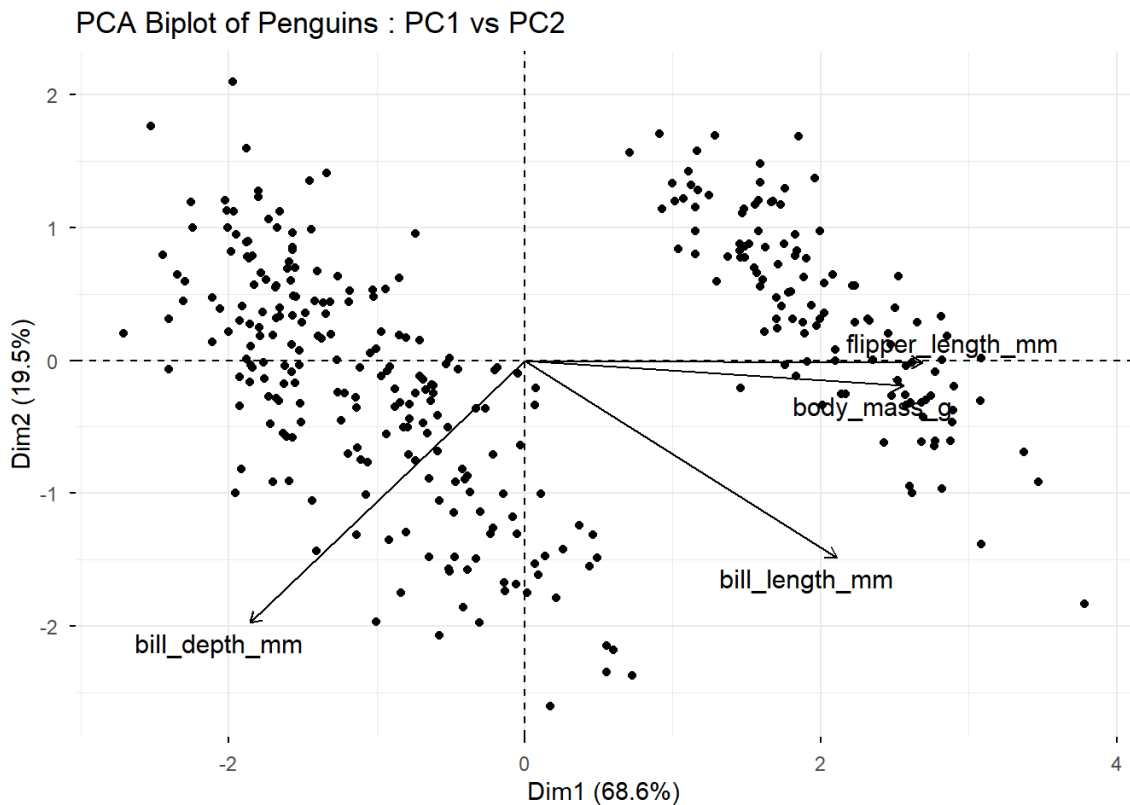
```
# Scree plot to visualize explained variance
fviz_eig(pca_result, addlabels = TRUE, main = "Scree Plot : PCA Explained Variance")
```



```
# Loadings of the first two principal components
pca_result$rotation[, 1:2]
```

```
##               PC1      PC2
## bill_length_mm  0.4537532 -0.60019490
## bill_depth_mm  -0.3990472 -0.79616951
## flipper_length_mm 0.5768250 -0.00578817
## body_mass_g      0.5496747 -0.07646366
```

```
# Biplot of the first two principal components
fviz_pca_biplot(pca_result, label = "var", habillage = "none", addEllipses = FALSE, col.var = "black", repel = TRUE, title = "PCA Biplot of Penguins : PC1 vs PC2")
```



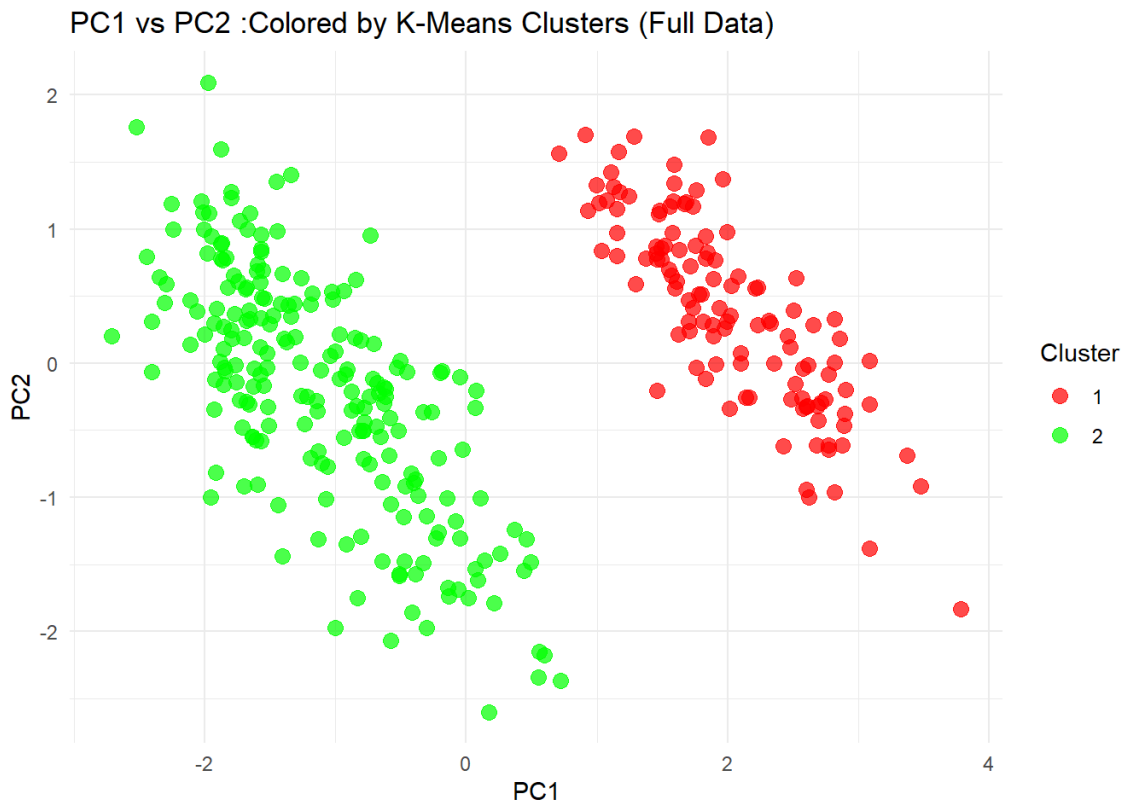
Task 7 : Combining K -means and PCA

```
set.seed(2026)
# Project data onto the first two principal components
pca_scores <- as.data.frame(pca_result$x[, 1:2])
colnames(pca_scores) <- c("PC1", "PC2")

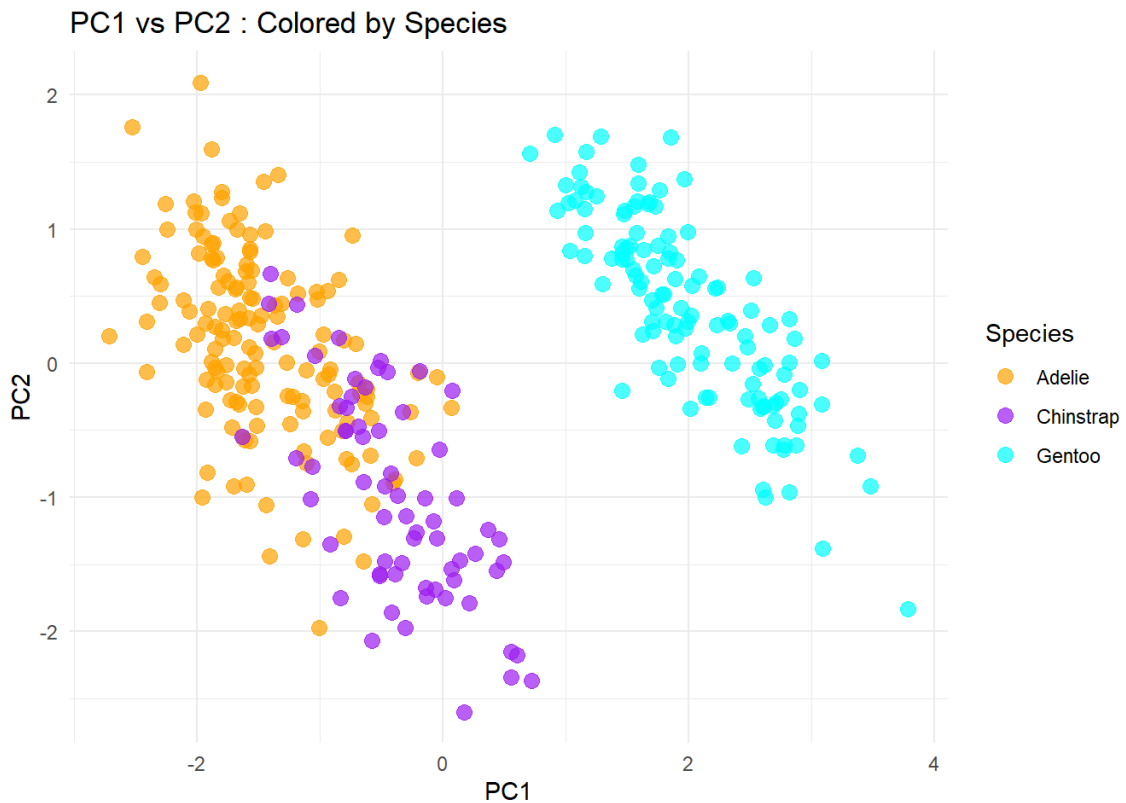
#K-means on PCA- reduced data
kmeans_pca <- kmeans(pca_scores, centers = k_opt, nstart = 25)
# Compare cluster assignments from PCA -reduced data on original4-dimensional data using confusion matrix
table(FullSpace = kmeans_model$cluster, PCspace = kmeans_pca$cluster)
```

```
##          PCspace
## FullSpace  1    2
##           1 119   0
##           2   0 214
```

```
# Scatterplot of PC1 vs PC2 colored by K-Means clusters(full data)
pca_scores$Cluster <- as.factor(kmeans_model$cluster)
ggplot(pca_scores, aes(x = PC1, y = PC2, color = Cluster)) +
  geom_point(size = 3, alpha = 0.7) +
  labs(title = "PC1 vs PC2 :Colored by K-Means Clusters (Full Data)")
+
  theme_minimal() +
  scale_color_manual(values = c("red", "green", "blue"))
```



```
# Scatterplot of PC1 vs PC2 colored by species
pca_scores$Species <- penguins_clean$species
ggplot(pca_scores, aes(x = PC1, y = PC2, color = Species)) +
  geom_point(size = 3, alpha = 0.7) +
  labs(title = "PC1 vs PC2 : Colored by Species") +
  theme_minimal() +
  scale_color_manual(values = c("orange", "purple", "cyan"))
```

Task 8 :Cluster validation and Quality Assessment

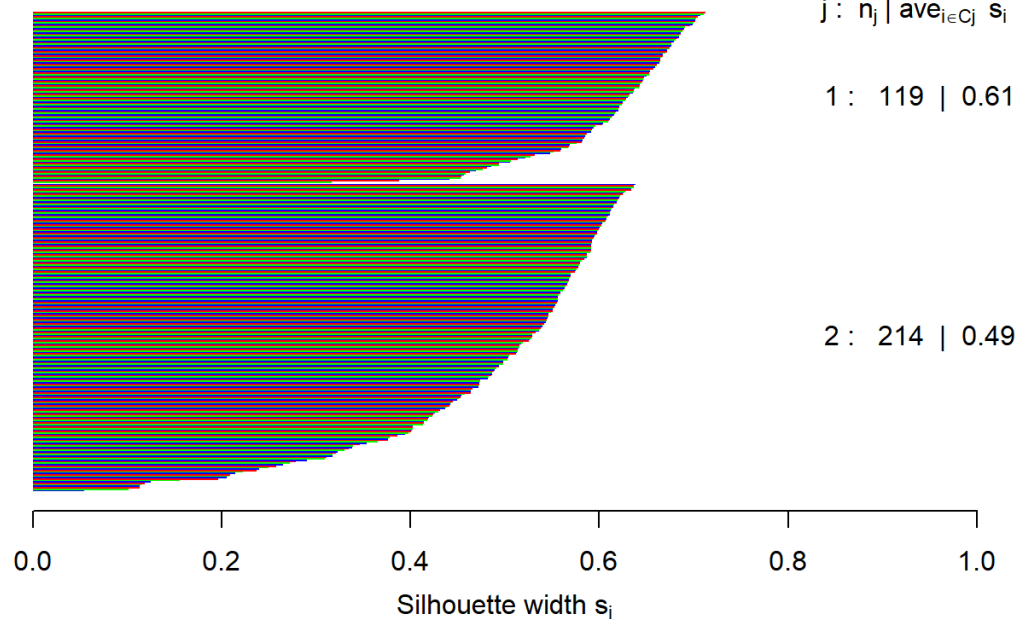
```
# Silhouette analysis for K-Means clusters
silhouette_kmeans <- silhouette(kmeans_model$cluster, dist(penguins_scaled))
mean_silhouette_kmeans <- mean(silhouette_kmeans[, 3])
mean_silhouette_kmeans
```

```
## [1] 0.5308174
```

```
plot(silhouette_kmeans, main = "Silhouette Plot for K-Means Clusters",
col = c("red", "green", "blue"), border = NA)
```

Silhouette Plot for K-Means Clusters

n = 333



```
# Dunn index for K-Means cluster
library(clValid)
dunn_index <- dunn(dist(penguins_scaled), kmeans_model$cluster)
dunn_index
```

```
## [1] 0.2729463
```

```
# Calinski-Harabasz Index for K-Means clusters
library(clusterSim)
```

```
## Loading required package: MASS
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```
ch_index <- index.G1(penguins_scaled, kmeans_model$cluster)
ch_index
```

```
## [1] 466.7476
```

```
# Comparing CH values for k= 2, 3, 4
ch_values <- numeric()

for (k in 2:4) {
  km <- kmeans(penguins_scaled, centers = k, nstart = 25)
  ch_values[k] <- index.G1(penguins_scaled, km$cluster)
}

ch_values
```

```
## [1]      NA 466.7476 427.7726 387.3515
```

Task 9 : Interpretation and Biological Insights

```
#Contingency table of K-Means clusters vs species
table(penguins_clean$Cluster, penguins_clean$species)
```

```
##
##      Adelie Chinstrap Gentoo
##  1         0           0    119
##  2      146          68       0
```

```
# PCA loadings Interpretation
loadings_df <- as.data.frame(pca_result$rotation[, 1:2])
loadings_df$Variable <- rownames(loadings_df)
print(loadings_df)
```

```
##
##      PC1      PC2      Variable
## bill_length_mm  0.4537532 -0.60019490 bill_length_mm
## bill_depth_mm -0.3990472 -0.79616951 bill_depth_mm
## flipper_length_mm 0.5768250 -0.00578817 flipper_length_mm
## body_mass_g      0.5496747 -0.07646366 body_mass_g
```